Joint Modeling of Longitudinal Biomarkers and Renal Graft Failure

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STAT 293 Final Project

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Background

- Kidney transplant patients are monitored repeatedly using several biomarkers:
  - Glomerular filtration rate (GFR)
  - Haematocrit
  - Proteinuria
- Over time, may be associated with renal graft failure.
- Repeated measurements are irregular, correlated, and subject to missingness.
- Standard survival models do not fully capture time-varying biomarker trajectories.
- **Joint longitudinal–survival modeling** provides a natural framework to study these processes simultaneously.

Research Questions

Primary objective

- Assess the strength and significance of the association between patient-specific longitudinal biomarker trajectories and time-to-renal graft failure.

Secondary objective

- Determine which biomarker contributes most strongly to graft failure risk after accounting for the others.

Joint modeling Approach

- Biomarkers are repeatedly measured over time.
- Biomarker processes are correlated.
- Dropout may be related to the failure process.

Data Structure

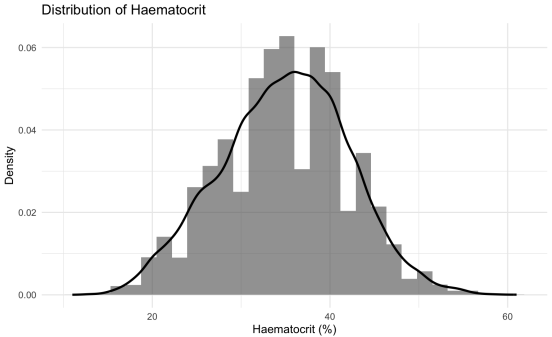
- Study population: **407** renal transplant patients
- Number of failures: **126** ($\approx 31\%$)
- Median follow-up: **11.9 years**

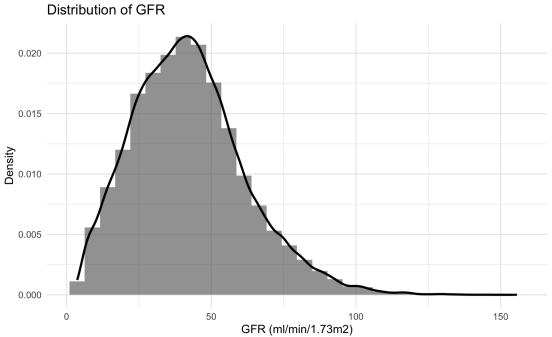
Available data sets

Data set	Outcome type	Size
`gfr`	Continuous	$70{,}514 \times 8$
`haem`	Continuous	$58{,}531 \times 8$
`prot`	Binary	$53{,}144 \times 8$
`surv`	Survival	407×6

- Baseline covariates include age, weight, gender, follow-up time, and failure indicator.

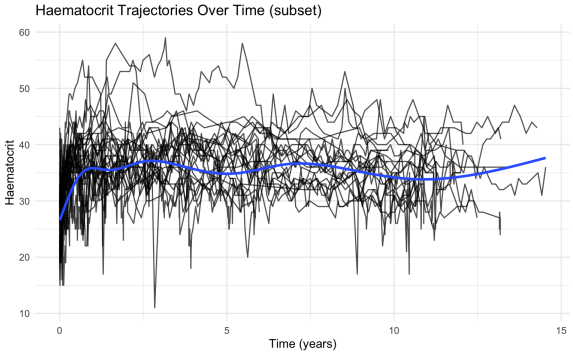
EDA (Normality)

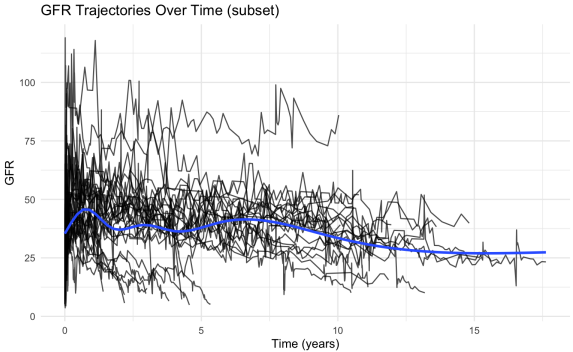




- Possible approach: Log transformation
  - Mild Skewness
  - Complicate interpretation

EDA (Non-Linearity & Heterogeneity)

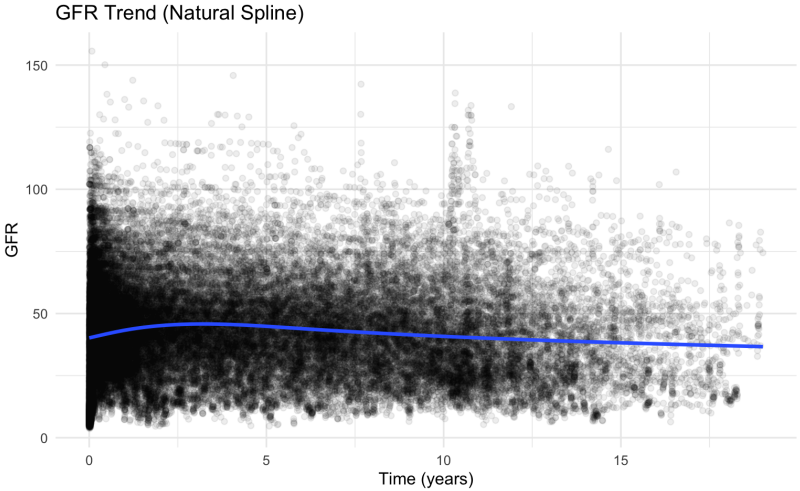




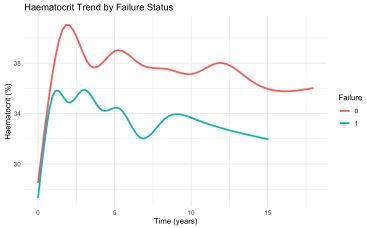
- Between-subject heterogeneity
  - in both baseline levels and slopes
  - random intercepts and slopes
- Mild Nonlinearity in GFR
  - Mixed-effects models with flexible time effects for GFR.

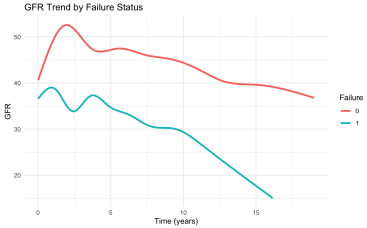
EDA (Non-Linearity)

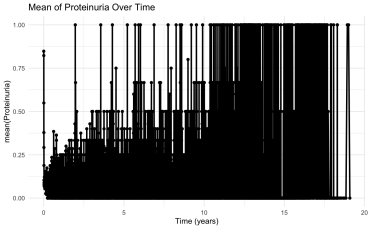
After adding a natural spline



EDA (Time-dependent association)

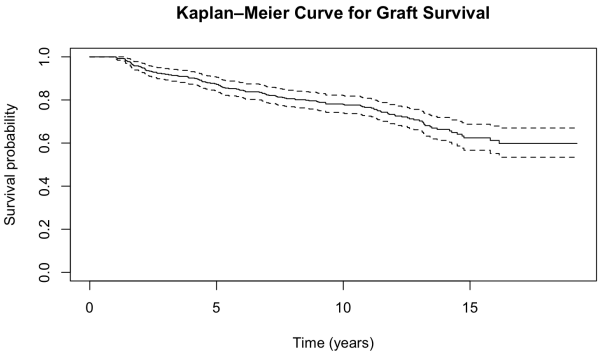






- Smooth trajectory curves by failure status
  - Lower Haematocrit and GFR among subjects experienced graft failure
- Mean Proteinuria Over Years
  - dynamic changes in kidney function indicators
- Time-dependent association between biomarker trajectories and graft failure risk.

EDA (Graft Survival)



- Graft failure risk evolves over time
  - modeling the hazard is meaningful.
- Events occur throughout follow-up.
  - time-to-event model

Missing Data and Estimation

Missingness in Biomarkers

- Approximately **27%** of longitudinal biomarker measurements were missing.
- Missingness depended on (by Logistic Regression):
  - prior biomarker values
  - graft failure status

- **MCAR is unlikely**
- **MAR assumption is plausible** (although MNAR cannot be ruled out)

Estimation Approach

- The joint model was fitted under a **maximum likelihood framework**.
- A **complete-case longitudinal dataset** was used for analysis.

Longitudinal Submodels

GFR model

$$\text{GFR}_{ij} = \beta_0^{(g)} + f(t_{ij}) + b_{0i}^{(g)} + b_{1i}^{(g)} t_{ij} + \varepsilon_{ij}^{(g)}$$

Haematocrit model

$$\text{Haem}_{ij} = \beta_0^{(h)} + \beta_1^{(h)} t_{ij} + b_{0i}^{(h)} + b_{1i}^{(h)} t_{ij} + \varepsilon_{ij}^{(h)}$$

Proteinuria model

$$\text{logit}\{P(\text{Prot}_{ij} = 1)\} = \beta_0^{(p)} + \beta_1^{(p)} t_{ij} + b_{0i}^{(p)}$$

- t_{ij} : time since transplant for subject i at visit j
- $f(t)$: natural cubic spline function of time
- b_{0i}, b_{1i} : subject-specific random intercepts and slopes
- $\varepsilon_{ij} \sim N(0, \sigma^2)$: measurement error

Survival Submodel

We model **time-to-graft failure** using Cox proportional hazards model

$$h_i(t) = h_0(t) \exp\left\{\gamma^\top \boldsymbol{X}_i + \alpha_1 m_{i,\text{gfr}}(t) + \alpha_2 m_{i,\text{haem}}(t) + \alpha_3 m_{i,\text{prot}}(t)\right\}.$$

- $\boldsymbol{X}_i$: baseline covariates (age, weight, gender)
- $m_{i,\text{gfr}}(t), m_{i,\text{haem}}(t), m_{i,\text{prot}}(t)$: current predicted biomarker values
- $\alpha_1, \alpha_2, \alpha_3$: association parameters linking biomarker trajectories to graft failure risk

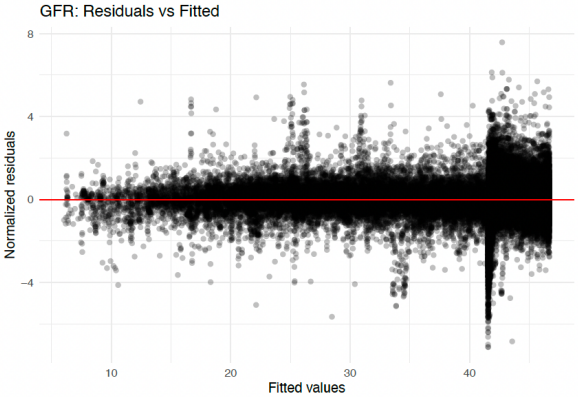
Interpretation

- The joint model quantifies how current biomarker status affects the hazard of graft failure.

Diagnostics for Continuous Longitudinal Outcome (GFR/Haematocrit)

Residual diagnostics

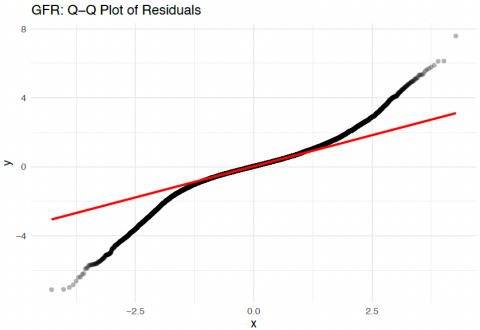
- Residuals vs fitted values were inspected to assess model fit.
- No strong systematic patterns were observed.

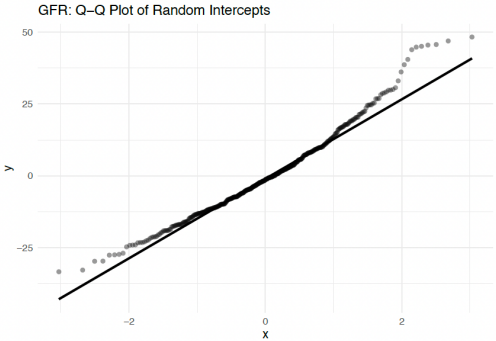


Diagnostics for Continuous Longitudinal Outcome (GFR/Haematocrit)

Normality checks

- Histogram and Q–Q plots suggest approximate normality of residuals.
- Q–Q plots of random intercepts support the normality assumption for random effects.

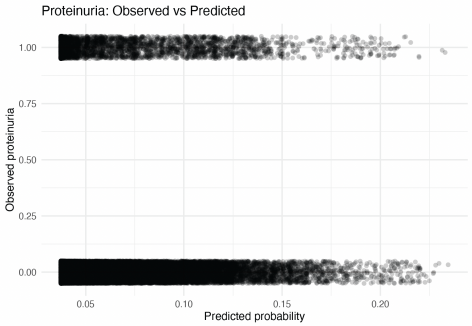




Diagnostics for Binary Longitudinal Outcome (Proteinuria)

Model calibration

- Predicted probabilities from the mixed logistic model were examined.
- Observed outcomes were compared with predicted probabilities.

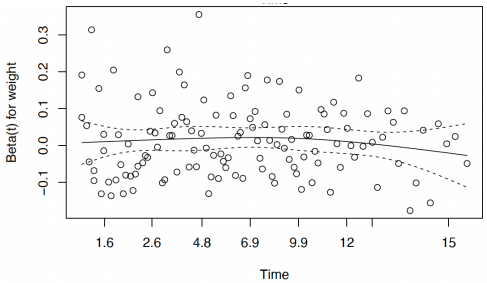


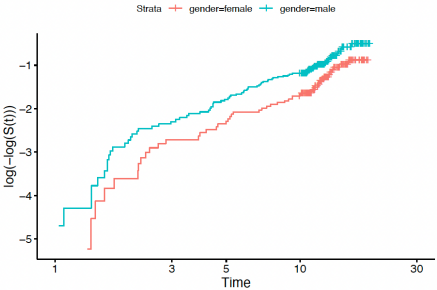
- Predictions are consistent with the observed binary outcomes.

Survival Model Diagnostics

Proportional hazards assumption

- Schoenfeld residual tests were performed.





- The global test was not significant ($p = 0.38$).
- Log(-log) survival curves were approximately parallel.

Conclusion

- No strong evidence of violation of the PH assumption.

Diagnostics Summary

- Residual diagnostics suggest acceptable fit for the longitudinal submodels.
- Random effects appear approximately normally distributed.
- Predicted probabilities from the proteinuria model align with observed outcomes.
- Schoenfeld residual tests and graphical checks suggest that the proportional hazards assumption holds.

Overall conclusion

- Model assumptions appear reasonable for the joint modeling framework.

Longitudinal Submodel Results

Outcome	Term	Estimate	SE	Statistic	P-value
GFR	Intercept	41.546	0.739	56.25	< 0.001
GFR	Spline(years) 1	-16.066	1.874	-8.57	< 0.001
GFR	Spline(years) 2	-18.468	3.886	-4.75	< 0.001
GFR	Spline(years) 3	-39.860	4.243	-9.39	< 0.001
Haematocrit	Intercept	33.238	0.202	164.60	< 0.001
Haematocrit	Years since transplant	0.659	0.039	16.97	< 0.001
Proteinuria	Intercept	-3.249	0.086	-37.60	< 0.001
Proteinuria	Years since transplant	0.108	0.005	22.69	< 0.001

- GFR changes nonlinearly over time.
- Haematocrit increases with time, while the probability of proteinuria also increases.

Joint Survival Model Results

Term	Estimate	SE	2.5%	97.5%	P-value
Age	-0.048	0.013	-0.074	-0.023	< 0.001
Weight	0.056	0.012	0.033	0.080	< 0.001
Male	0.389	0.233	-0.081	0.847	0.102
Current GFR	-0.154	0.010	-0.174	-0.135	< 0.001
Current Haematocrit	0.127	0.025	0.079	0.176	< 0.001
Current Proteinuria	0.226	0.076	0.078	0.376	0.004

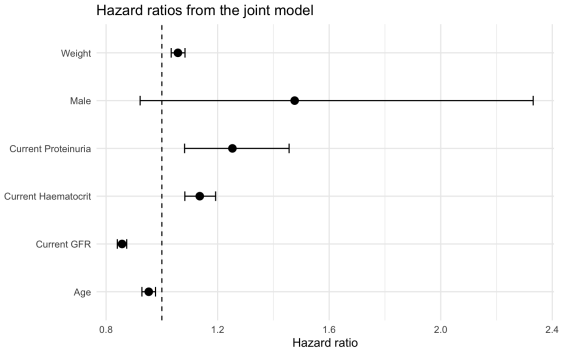
- Lower current GFR is associated with higher graft failure risk.
- Higher current haematocrit and proteinuria are associated with higher risk.

Hazard Ratios from the Joint Model

Covariate	HR	95% CI	P-value
Age	0.953	(0.928, 0.977)	< 0.001
Weight	1.058	(1.034, 1.083)	< 0.001
Male	1.476	(0.922, 2.332)	0.102
Current GFR	0.857	(0.840, 0.874)	< 0.001
Current Haematocrit	1.136	(1.082, 1.193)	< 0.001
Current Proteinuria	1.253	(1.081, 1.456)	0.004

- Hazard ratios are obtained by exponentiating the joint survival model coefficients.
- Current biomarker values represent the predicted trajectories from the longitudinal models.

Which Biomarker Contributes Most to Graft Failure Risk?



Secondary analysis:

- Among the biomarkers, **current proteinuria** shows the largest harmful association with graft failure.
- **Current GFR** shows a protective association (HR < 1).

Interpretation of Main Findings

- **GFR:** Higher current GFR is associated with lower hazard of graft failure.
- **Proteinuria:** Higher current proteinuria is associated with increased hazard.
- **Haematocrit:** Also significantly associated with failure risk after adjustment.
- Among the three biomarkers, **proteinuria appears to be the strongest harmful risk indicator** based on hazard ratio magnitude.

Sensitivity Analysis for Possible MNAR Departures

- MAR mechanism applied in analysis
- Consider possible **MNAR departures**
  - a simple δ-shift sensitivity analysis for robustness.
- We decreased post-dropout GFR values by:
  - $\delta = 5$
  - $\delta = 10$
- The joint model was then refit under these perturbed scenarios.

Purpose

- Evaluate whether the estimated association between GFR and graft failure is robust to plausible worsening of unobserved GFR values.

Result

- Estimated association between current GFR and graft failure remains **unchanged**
- Highly robust to plausible deviations from the MAR assumption in the longitudinal biomarker process.

Conclusions

- Longitudinal biomarker trajectories are significantly associated with renal graft failure.
- Lower GFR is strongly associated with increased graft failure risk.
- Proteinuria is an important adverse biomarker and appears to have the strongest harmful contribution.
- Multivariate joint modeling provides a principled way to account for:
  - repeated measures,
  - cross-biomarker dependence

Thank you!

Questions?